

ZHUOJUN HAN (韩卓骏)

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🎓 EDUCATION

Fudan University (FDU), Shanghai, China

2023 – Present

Undergraduate student in Microelectronics Engineering, expected March 2026

👤 EXPERIENCE

Research & Engineering Projects

2024 – Present

CIM, VLM, Hardware-Algorithm Co-design State Key Laboratory of Integrated Chips and Systems (SKLICS)

- **CIM-Pruner (IEEE ISCAS 2026):** a hardware-algorithm co-design framework for efficient VLM inference, featuring intra-chunk token pruning/merging, early-quit bitonic sorting, and a dual-mode token processing unit; implemented in 28nm CMOS, achieving 25.64 TOPS/W (@200MHz, INT8)
- **12nm CIM Accelerator Tape-out:** Designed and taped out a 12nm CIM-based accelerator **VGCIM** for efficient and temporally consistent video generation; completed full ASIC backend flow using Synopsys tools (DC, ICC2 for P&R, FM for equivalence checking, VCS for simulation), including synthesis, floorplanning, routing, and verification; incorporated token-level pruning/merging, approximate popcount-based frame selection, and in-memory non-linear acceleration (Softmax/LayerNorm) to optimize compute and memory efficiency
- **CSIRO Kaggle Competition (Top 7%):** Built an end-to-end pipeline for image-to-biomass prediction, leveraging DINOv3 and SigLIP for efficient inference, and applying feature engineering, ensemble learning, and cross-validation; achieved top 7% ranking in the Kaggle CSIRO - Image2Biomass Prediction competition
- **Ternary LLM CIM Accelerator (DAC 2027 Submission):** Designed a compute-in-memory accelerator for ternary quantized LLMs, focusing on efficient encoding and MAC optimization; manuscript under preparation for ACM DAC 2027. Code: <https://github.com/LArielOzjH/TernaryCIM>
- **On the Difficulty of Robotic Arm Manipulation Using Imitation Learning and Offline Reinforcement Learning:** Designed a robotic manipulation task (PandaPickPlaceBread) in Robosuite (MuJoCo), collected expert demonstrations via teleoperation, and trained BC (MLP), sequence-based BC (GRU), and offline RL (BCQ) policies; conducted controlled experiments evaluating success rate, stability, and robustness, identifying key factors affecting policy performance and reliability. Code: <https://github.com/LArielOzjH/robotic-manipulation-il-offline-rl.git>

⚙️ SKILLS

- **Hardware & EDA:** SystemVerilog, ASIC design flow (Synopsys DC, ICC2, PrimeTime, Formality), VCS
- **Machine Learning & AI:** PyTorch, NumPy, scikit-learn, feature engineering, ensemble learning
- **Systems & Tools:** Linux, Git, Zsh, SSH, CUDA (basic)
- **Programming:** Python, C++, TCL

📖 MISCELLANEOUS

- Languages: English (Fluent), Mandarin (Native)
- Research Interests: AI accelerators, world models, video generation, hardware-software co-design